LLM Evaluation Case Study

Prompt Engineering & RLHF Portfolio

Case ID: [004] | Date: October 13, 2026

1. The Prompt

User Input:

Guess the next number: 0, 74, 1, -1, 2, -1, -1, 1, -1, -1, 0, 0, -1, 0, 1, -2, 0, 0, -3, 0, ?

Evaluator's Hidden Context / Ground Truth:
These are the first terms (for $n = 0, 1, 2, \ldots$) of the sequence defined by the function: $a[n] = \lfloor \tan(\tan(n)) \rfloor$.

2. Model Responses

Model A (gpt-6-luna-max)

There's no uniquely determined next number without a rule. If I had to make a casual guess, I'd say **0**—but many other answers fit the same list.

Model B (qwen3.8-27b)

To determine the next number in the sequence, let's analyze the pattern by splitting the sequence into two interleaved subsequences: one for the odd positions and one for the even positions. The sequence is: Index: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21
Value: 0, 74, 1, -1, 2, -1, -1, 1, -1, -1, 0, 0, -1, 0, 1, -2, 0, 0, -3, 0, ?
Step 1: Analyze the even-positioned terms (Indices 2, 4, 6, ..., 20)
Let's look at the values at indices 2, 4, 6, 8, 10, 12, 14, 16, 18, 20:
- Index 2: 74
- Index 4: -1
- Index 6: -1
- Index 8: 1
- Index 10: -1

- Index 12: 0
- Index 14: 0
- Index 16: -2
- Index 18: 0
- Index 20: 0

This subsequence (74, -1, -1, 1, -1, 0, 0, -2, 0, 0) does not show an obvious simple arithmetic or geometric pattern, especially with the outlier ”74” at the start. Let’s look at the odd-positioned terms first, as they might provide a clearer clue.

Step 2: Analyze the odd-positioned terms (Indices 1, 3, 5, ..., 21)

Let’s look at the values at indices 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21:
- Index 1: 0
- Index 3: 1
- Index 5: 2
- Index 7: -1
- Index 9: -1
- Index 11: 0
- Index 13: -1
- Index 15: 1
- Index 17: 0
- Index 19: -3
- Index 21: ?

This also looks somewhat irregular. Let’s re-examine the entire sequence for a different grouping.

Step 3: Look for a pattern in groups of three or four

Let’s try grouping by 3:
(0, 74, 1) (-1, 2, -1) (-1, 1, -1) (-1, 0, 0) (-1, 0, 1) (-2, 0, 0) (-3, 0, ?)

Let’s look at the first element of each triplet: 0, -1, -1, -1, -1, -2, -3. This looks like a sequence that stays at -1 for a while, then decreases.
Differences are: -1, 0, 0, 0, -1, -1. This doesn’t seem perfectly consistent yet.

Let’s look at the second element of each triplet: 74, 2, 1, 0, 0, 0, 0.
This is a decreasing sequence. It stabilizes at 0.

Let’s look at the third element of each triplet: 1, -1, -1, 0, 1, 0, ?
Let’s re-evaluate the grouping. Maybe the ”74” is a typo for something else, or part of a different pattern. Let’s look at the sequence again. Notice the pairs:
(0, 74) (1, -1) (2, -1) (-1, 1) (-1, -1) (0, 0) (-1, 0) (1, -2) (0, 0) (-3, 0) (?, ?) $\rightarrow$ No, the question asks for the 21st term. Let’s try grouping by **triplets** again, but looking at the columns.

Col 1 (Indices 1, 4, 7, 10, 13, 16, 19): 0, -1, -1, -1, -1, -2, -3
Col 2 (Indices 2, 5, 8, 11, 14, 17, 20): 74, 2, 1, 0, 0, 0, 0
Col 3 (Indices 3, 6, 9, 12, 15, 18, 21): 1, -1, -1, 0, 1, 0, ?

Let’s analyze **Column 1**: $0, -1, -1, -1, -1, -2, -3$
The values are: $n = 1 : 0$, $n = 2 : -1$, $n = 3 : -1$, $n = 4 : -1$, $n = 5 : -1$, $n = 6 : -2$, $n = 7 : -3$.

Let’s analyze **Column 2**: $74, 2, 1, 0, 0, 0, 0$
This drops quickly to 0 and stays there. This suggests the ”74” might be an anomaly or part of a specific rule like $\lfloor \frac{100}{n} \rfloor$? No.

Let’s analyze **Column 3**: $1, -1, -1, 0, 1, 0, ?$
Let $C1_i, C2_i, C3_i$ be the i-th triplet. Is there a relationship between $C1$ and $C3$?
Let’s reconsider the **entire sequence** as a single stream. What if the sequence is related to **digits**? No.

Let’s try **interleaving** again, but with a different split. Maybe the sequence is composed of two sequences, A and B.
A (odd indices): 0, 1, 2, -1, -1, 0, -1, 1, 0, -3, ?

B (even indices): 74, -1, -1, 1, -1, 0, 0, -2, 0, 0
Let's break A into chunks of 3:
(0, 1, 2) (-1, -1, 0) (-1, 1, 0) (-3, ?, ?)
Is there a pattern in the second element of each chunk? 1, -1, 1, ?
Alternating 1, -1, 1... Next should be **-1**.
If the pattern is an alternating sign: $(-1)^{k+1}$, therefore, the next number is **-1**.
Answer: **-1**

3. Evaluation & Reasoning

Decision: Tie (Both Failed)
Category/Domain: Mathematics / Logical Reasoning & Pattern Recognition

Detailed Justification

- **Instruction Following & Accuracy:** Both models attempted to answer the prompt by guessing the next number, but neither successfully identified the true mathematical function generating the sequence ($a[n] = \lfloor \tan(\tan(n)) \rfloor$). Thus, both failed the primary objective of the prompt.
- **Hallucinations of Logic:** Model A correctly identified the absence of an obvious simple rule and made an honest, declared guess. Model B, conversely, suffered from severe "hallucination of logic". It generated a massive, exhaustive analysis—testing groupings by twos, threes, columns, and interleaved sequences—and ultimately fabricated a non-existent alternating pattern ("1, -1, 1...") simply to forcefully justify a definitive answer.
- **Verbosity Bias:** Model B exhibited extreme verbosity bias. It wasted significant computational tokens on a rambling, arbitrary chain of thought that led to a flawed conclusion. Model A was concise and direct.
- **Conclusion:** The result is evaluated as a **Tie**, as neither model solved the mathematical problem. However, in a real-world scenario, Model A's behavior is inherently safer and preferable. Acknowledging a lack of information is vastly superior to confidently generating and defending a fabricated logical proof, which can severely mislead the user.